**Unit:** Unit 0: Foundations of Algebra
**Topic:** GCF LCM

**Name:** ________________
**Date:** __________

# Foundation (Jalapeno)

**Q1.** What is the GCF of 12 and 18?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 9

**Q2.** A space probe travels 30 miles in 5 minutes. How many miles will it travel in 15 minutes if it maintains the same GCF of speed?
(A) 60 miles
(B) 90 miles
(C) 120 miles
(D) 150 miles
(E) 180 miles

**Q3.** What is the LCM of 8 and 12 using prime factorization?
(A) 24
(B) 32
(C) 40
(D) 48
(E) 56

**Q4.** A lab experiment requires 15 ml of a chemical. If the chemical is only available in bottles of 3 ml or 5 ml, what is the LCM of the bottle sizes?
(A) 15 ml
(B) 30 ml
(C) 45 ml
(D) 60 ml
(E) 75 ml

**Q5.** What is the GCF of 24 and 30 using factors?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 12

**Q6.** A space station has two modules with 18 and 24 solar panels. What is the GCF of the number of solar panels?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 9

**Q7.** A physics experiment involves a pendulum with a length of 12 inches and a swing period of 8 seconds. What is the LCM of the length and period?
(A) 24
(B) 32
(C) 40
(D) 48
(E) 56

**Q8.** What is the LCM of 9 and 12 using multiples?
(A) 18
(B) 27
(C) 36
(D) 45
(E) 54

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**Open Ended Questions (FRQ)**

**Q9.** A lab requires 12 liters of a chemical to conduct an experiment. If the chemical is only available in bottles of 4 liters or 6 liters, determine the number of each type of bottle needed to obtain exactly 12 liters using GCF and LCM.

**Q10.** A space mission requires a spacecraft to travel to two planets with orbital periods of 15 days and 20 days. Determine the LCM of the orbital periods and explain how this value is used to plan the mission.

**Chef's Tips**



- Emphasize the importance of prime factorization in finding GCF and LCM.
- Use real-world examples to illustrate the application of GCF and LCM in science and space exploration.
- Encourage students to use diagrams and models to visualize the concepts of GCF and LCM.